# INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

## Subject Code : 23A0360Se

## UNIT II

### Electric Vehicle Power Plant and Drives

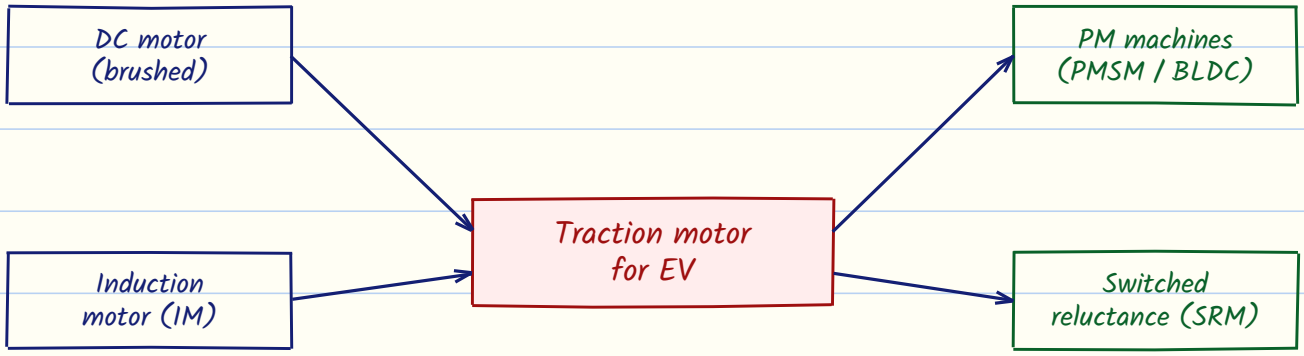
## I. Introduction - Electric Vehicle Power Plants

The **power plant** of an EV is the electric **propulsion system**: battery -> power electronic converter -> electric motor -> transmission. Here the "engine" is the **electric machine plus its power converter and controller**.

### Requirements of an EV traction motor

- High torque at low speed (starting, hill climbing) and high power at high speed - wide constant-power range (3-4 times base speed).
- High **efficiency** over a wide torque-speed range, especially at light loads (drive cycles).
- High power density and torque density (small size, low weight), good cooling.
- Regenerative braking capability, fast torque response, robustness, low noise, low cost, high reliability.

## Types of electric machines used




Early EVs used brushed **DC motors** (simple control but brushes need maintenance). Modern EVs use brushless machines: **induction**, **permanent magnet (PM)** and **switched reluctance (SRM)** motors with inverter control.

## 2. Induction Machines (IM)

The most widely used machine in EVs (e.g. Tesla Model S early versions). It has a stator with three-phase winding and a **squirrel-cage rotor** (bars short-circuited by end rings).

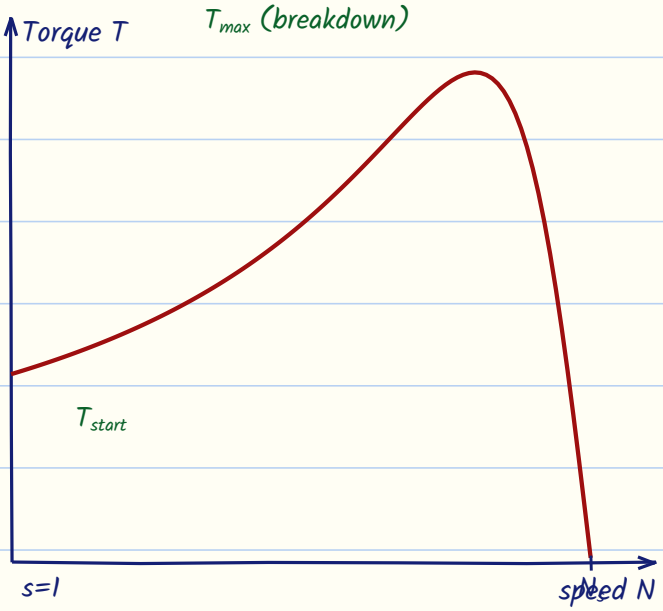
### Working principle

- The stator three-phase currents create a **rotating magnetic field** at synchronous speed $N_s = 120\ f / p$ (rpm).
- The field cuts the rotor conductors and induces emf and current in them (transformer action). Torque is produced

by interaction of the rotor current with the air-gap flux.

- The rotor runs slightly below $N_s$. The difference is called **slip**: $s = (N_s - N) / N_s$. Torque exists only when $s \neq 0$.

$$T = (3\,p / 2\,\omega_s) \cdot V^2 (R_2/s) / [\,(R_1 + R_2/s)^2 + (X_1 + X_2)^2\,]$$




Slip $s = (N_s - N)/N_s$

Sync speed $N_s = 120\,f / p$

Stable region: between $T_{max}$ and $N_s$

Speed control: V/f constant shifts the curve sideways.

## Control of IM in EVs

- **Constant V/f control** (scalar) - keeps flux constant below base speed. Simple but poorer dynamic response.
- **Field-oriented (vector) control, FOC** - stator current split into flux-producing ($i_d$) and torque-producing ($i_q$) components controlled independently like a DC motor; **field weakening** above base speed gives constant-power region.
- **Direct torque control (DTC)** - directly controls torque and flux with hysteresis controllers; fast response.

### Merits and demerits

- + Simple, rugged, low cost, no magnets, low maintenance, good field-weakening range, reliable at high speed.
- - Lower efficiency than PM motors (rotor copper loss), lower power factor, rotor heating, control more complex, lower power density.

## 3. Permanent Magnet (PM) Machines

Magnets (NdFeB, SmCo) in the rotor give excitation flux without field current, hence no rotor copper loss - **highest efficiency and power density** among EV motors. Used in Toyota Prius, Nissan Leaf, Chevrolet Volt.

### Types

- **PM synchronous motor (PMSM)** - sinusoidal back-emf, sinusoidal stator current; smooth torque, controlled by FOC.
- **PM brushless DC motor (BLDC)** - trapezoidal back-emf, rectangular current in two phases at a time using rotor position (Hall) sensors; simple control, some torque ripple.

### Rotor construction

- **Surface-mounted (SPM)** - magnets glued on rotor surface; low saliency, limited field weakening.
- **Interior (IPM)** - magnets buried inside rotor; gives

**reluctance torque** also: $T = (3p/2)[\psi_m i_q + (L_d - L_q) i_d i_q]$; wide field-weakening range; most popular in EVs.

### Merits and demerits

- + High efficiency, high torque / power density, good power factor, compact.
- - High magnet cost, magnets can demagnetise at high temperature, limited constant-power range, back-emf at high speed (needs field weakening current), uncontrolled generation fault risk.

## 4. Switched Reluctance Machines (SRM)

A **doubly-salient** machine: both stator and rotor have teeth (salient poles). The stator has concentrated coil windings, the rotor has **no winding and no magnets** - just laminated steel. Common types 6/4 (3-phase) and 8/6 (4-phase).

- Torque is produced by the tendency of the rotor to align with the energised stator pole to **minimise reluctance** (maximise inductance).
- Torque is independent of the direction of current: $T = \frac{1}{2} i^2 (dL/d\theta)$. Phases are excited in sequence, synchronised with rotor position (shaft encoder needed).
- + Very rugged and cheap, high-speed capability, fault tolerant (phases independent), wide constant-power

range, no magnet cost.

- – **Torque ripple**, acoustic noise and vibration, needs special converter (more wires), lower power density than PM.

| Property | DC | IM | PM | SRM |
|---|---|---|---|---|
| Efficiency | Low | Medium | Highest | Good |
| Power density | Low | Medium | Highest | Good |
| Cost | Medium | Low | High (magnets) | Lowest |
| Control | Simplest | Complex (FOC) | Complex | Moderate |
| Ruggedness / maint. | Brushes | Rugged | Good | Most rugged |
| Max speed | Low | High | Medium | Highest |

## 5. Power Electronic Converters – DC/DC Converters

A DC/DC converter changes the DC voltage level (battery to inverter, battery to auxiliary loads, fuel cell / ultracap interface). It uses a high-speed switch (MOSFET / IGBT) operated with duty ratio $D = t_{on} / T$.

### (a) Buck converter (step-down)

$$V_o = D\,V_{in} \quad (0 < D < 1)$$

When the switch S is ON, the inductor current rises and energy goes to the load; when OFF, the diode D freewheels

and L discharges. Output ripple is reduced by L and C. Used for 12 V / 48 V auxiliary supplies.

### (b) Boost converter (step-up)

$$V_o = V_{in} / (1 - D)$$

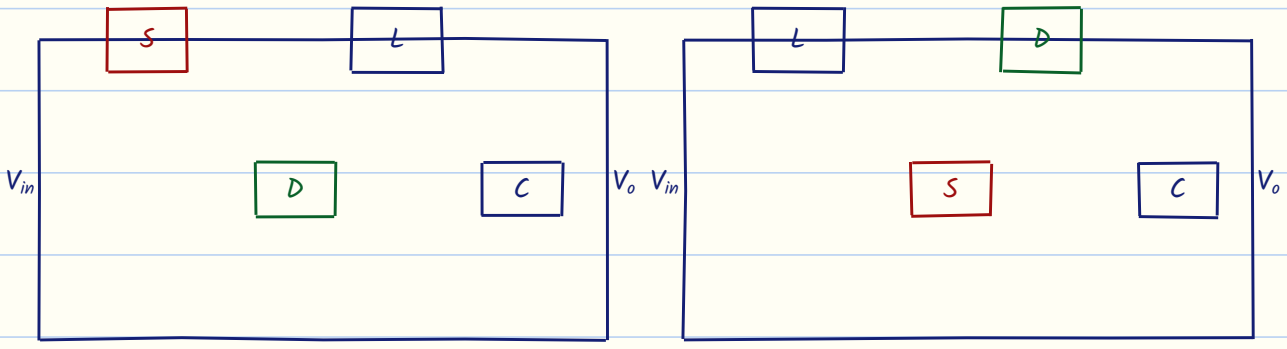
S ON: inductor stores energy from the source. S OFF: inductor emf adds to $V_{in}$ and charges the output capacitor through the diode. Used to raise battery voltage to the higher DC-link voltage of the inverter.

### (c) Buck-boost converter

$$V_o = - V_{in} D / (1 - D)$$

Gives output either lower ($D < 0.5$) or higher ($D > 0.5$) than input with **reversed polarity**. Inductor current: $\Delta I_L = V_{in} D T / L$ (CCM).

Buck (step-down) Boost (step-up)




$V_o = D V_{in}$ $V_o = V_{in}/(1-D)$

## (d) Isolated DC/DC converter

When **galvanic isolation** between battery and the vehicle chassis / other bus is needed (safety, large voltage ratio), a **high-frequency transformer** is used. Types: **flyback**, **forward**, **push-pull**, **half-bridge** and **full-bridge**. The DC is first chopped (inverted) at 20-100 kHz, stepped up/down by the transformer, then rectified and filtered.

Isolated DC/DC: galvanic isolation through high-frequency transformer

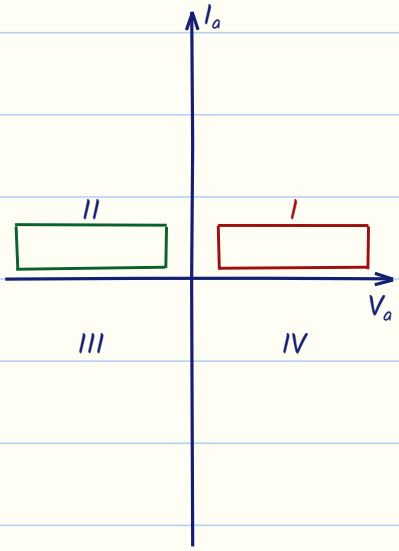



$V_o = D\,(N_2/N_1)\,V_{in}$ (forward / bridge type)

- Dual-active-bridge (DAB) is a bidirectional isolated converter - popular for onboard chargers and battery - DC link interface.
- Advantages of HF transformer: small size and weight, isolation, multiple outputs; losses: core and winding loss, leakage inductance spikes (need snubber).

# 6. Two-Quadrant Chopper and Switching Modes

A DC motor drive needs power flow both **from battery to motor (motoring)** and **motor to battery (regeneration)**. A **two-quadrant chopper (Class C)** has two switches $S_1$, $S_2$ with anti-parallel diodes $D_1$, $D_2$, forming a half-bridge leg. Terminal voltage stays positive and current may be positive or negative.




Quadrant I : forward motoring ($S_1$ chopping, buck mode)

Quadrant II : forward regeneration ($S_2$ chopping, boost mode)

Two-quadrant chopper: $V_a > 0$ always, $I_a$ can be + (motor) or − (regen).

## Operating modes

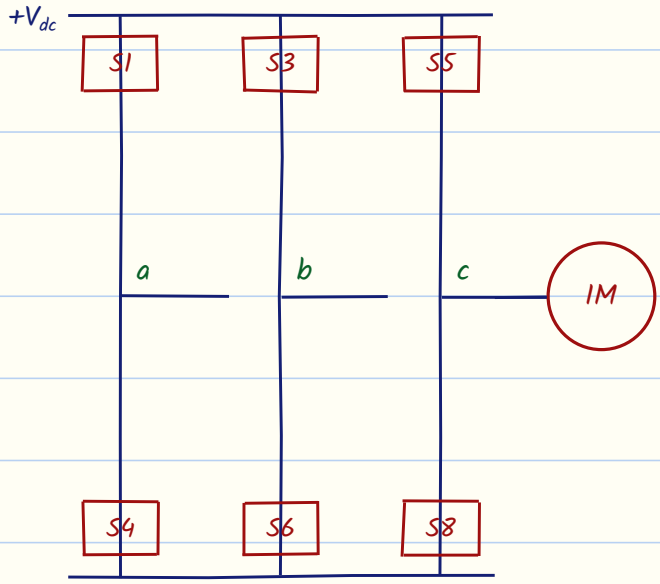
- **Motoring (buck mode)**: $S_1$ is switched with duty D, $S_2$ off. Mode 1: $S_1$ ON, current flows from battery through motor. Mode 2: $S_1$ OFF, $D_2$ freewheels. $V_a = D\ V_b$.
- **Regeneration (boost mode)**: $S_2$ is switched, $S_1$ off. Mode 1: $S_2$ ON, motor armature (back emf $E > V_b$-side) stores energy in inductance. Mode 2: $S_2$ OFF, current flows

through $D_1$ into the battery. $V_b = V_a/(1 - D)$.

- **Switching modes of control**: (i) constant-frequency **PWM** (vary $t_{on}$); (ii) variable-frequency (constant $t_{on}$); (iii) current-limit control. Dead time is provided between $S_1$ and $S_2$ to avoid shoot-through.

## 7. AC Drives - PWM and Current Control

An AC drive (IM or PMSM) is fed by a **three-phase voltage-source inverter (VSI)** from the DC bus. PWM produces variable-voltage variable-frequency (VVVF) output.



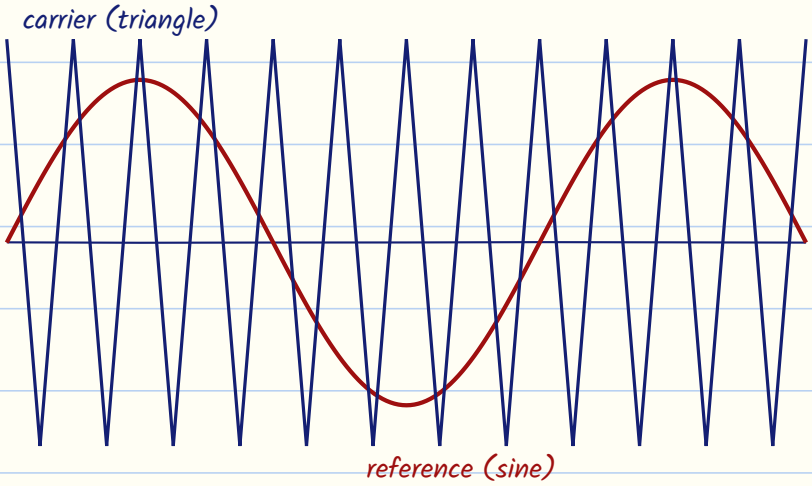

*Three-phase voltage source inverter*

*6 IGBT / MOSFET + diodes*
*PWM gives variable V and f*
*Feeds IM or PMSM from battery*
*Bidirectional: regen possible*

### (a) Sinusoidal PWM (SPWM)

A sinusoidal reference (modulating) wave at the required output frequency is compared with a high-frequency triangular carrier. Where the reference exceeds the carrier the upper switch is ON. Amplitude modulation index $m_a$ =

$V_{ref}/V_{carrier}$; carrier ratio $m_f = f_c/f_{ref}$. Fundamental line-line rms = 0.612 $m_a$ $V_{dc}$. Over-modulation ($m_a$ > 1) gives up to square wave with lower utilisation of harmonics.




Sine PWM:

$v_{ref}$ > $v_{tri}$ -> upper switch ON
$v_{ref}$ < $v_{tri}$ -> lower switch ON
$m_a = V_{ref}/V_{tri}$
$V_{LL,rms} = 0.612\ m_a\ V_{dc}$
(0 < $m_a$ <= 1, linear)

## (b) Space-vector PWM (SVPWM)

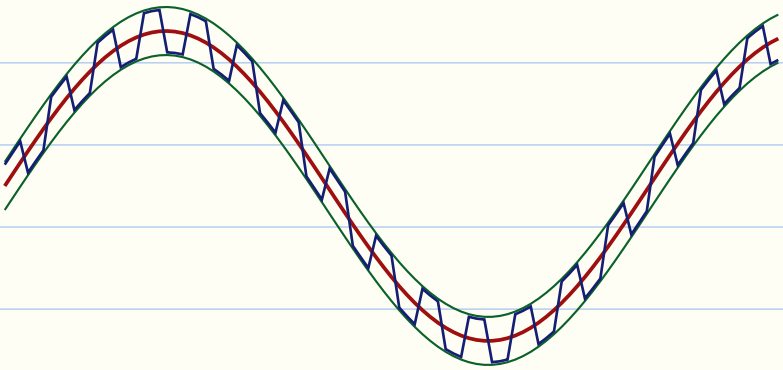
- Treats the three phase voltages as one rotating reference vector built from the 8 inverter switching states (6 active + 2 zero vectors).
- About **15 % higher DC bus utilisation** than SPWM and lower harmonic distortion, easy digital implementation - preferred in EVs.

## (c) Current control methods

- **Hysteresis (bang-bang)** current control - the phase current is kept within ±h of the reference by switching the inverter leg when the band is crossed. Fast, simple;

frequency not fixed.

- **Ramp-comparison (PWM) current control** – the current error passes through a PI controller and is compared with the carrier. Fixed frequency, better harmonic spectrum.
- **Synchronous-frame PI control** – currents transformed to d-q axes become DC quantities, controlled by PI regulators (used with FOC).



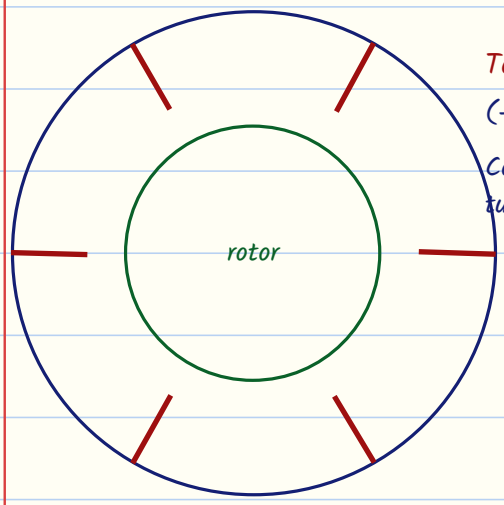
*Hysteresis current control*

*actual current i stays inside band $i_{ref} \pm h$ (green). Simple, fast, robust but switching frequency varies.*

## 8. Switched Reluctance Machine Drives

An SRM needs a converter that can energise each phase independently in the right rotor position. Common converter: **asymmetric half-bridge** per phase (two switches + two diodes). Switching both ON applies +V; both OFF applies -V (demagnetise); one OFF with diode conduction gives zero voltage (freewheel).

Phase winding with asymmetric half-bridge:




Torque $T = \frac{1}{2} i^2 (dL/d\theta)$

(+ve when $dL/d\theta > 0$ : rising L)

Control variables: turn-on angle $\theta_{on}$, turn-off angle $\theta_{off}$ and current / voltage.




6/4 SRM

## (a) Voltage control

- The phase voltage (average) is controlled by PWM duty ratio or by choosing **turn-on and turn-off angles** ($\theta_{on}$, $\theta_{off}$). At high speed the back emf is large, current is naturally limited and **single-pulse (angle) control** is used.
- Simple, but the current peak is not directly controlled, so torque ripple and peak currents are larger at low speed.

## (b) Current control

- Used at low speed and starting, where the current would otherwise rise too high. The phase current is regulated by **hysteresis** or PWM current controller around a reference $I_{ref}$, produced from the torque command (torque sharing between phases reduces ripple).
- Gives good torque control and protects switches; needs

current sensors in each phase.

- Overall: speed loop (outer) -> torque / current command -> current loop (inner) -> converter; rotor position feedback from encoder or sensorless methods.

Important questions: (1) Compare IM, PM and SRM for EV. (2) Buck and boost converter operation with equations. (3) Two-quadrant chopper. (4) Sine PWM and space vector PWM. (5) SRM voltage and current control.